YOLO26n for Infrared Small-UAV Detection: Experiments and Numerical Diagnosis

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***Abstract*—We evaluate YOLO26n, shallow pinwheel-shaped convolution (PConv), and a stride-4 P2 branch for infrared UAV detection. A duplicate-audited SIRST-UAVB holdout contains 1,438 training and 505 validation images. Historical baseline and PConv final AP50/AP50:95 were 56.2%/23.2% and 35.8%/14.1%, but seven and eleven validation-loss rows contained NaN. Early-EMA replay reproduced nonfinite AMP forward outputs; full FP32 replay was finite. A separate P2 experiment retained AMP training and used FP32 validation and checkpoint selection. All 92 epochs had finite histories; final AP50/AP50:95 were 47.405%/20.652%. P2 did not exceed the baseline FP32 revalidation. Different checkpoint-selection precision limits architecture-only comparisons.**

***Index Terms*—Infrared UAV detection, YOLO26n, small targets, numerical diagnosis, mixed precision.**

I. Background and Study Design

Small infrared UAVs contain few pixels and limited texture. Bright background structures and birds can resemble targets, while small coordinate errors strongly affect box overlap. We use YOLO26n as a lightweight bounding-box detector [1], [2]. Pinwheel-shaped convolution (PConv) provides directional shallow sampling [3]; a high-resolution P2 branch tests whether preserving finer spatial features helps detection.

B0 is the baseline, B1 replaces layer 1 with PConv, and B2 adds a stride-4 P2 branch. We report the original B0/B1 outcomes, a subsequent numerical diagnosis, and a separate P2 recovery experiment. The original comparison queue stopped before P2; the recovery experiment does not change its failed status or overwrite historical losses.

II. Dataset Audit and Preparation

SIRST-UAVB contains 3,000 grayscale images at 640 by 512 pixels [3]. Acquired native labels contain 2,961 UAV and 1,747 bird boxes, identified by cross-checking masks and the authors' annotation description. UAV is the single training class; bird boxes are retained for error analysis.

The original 2,400/600 split shares 496 exact-pixel image groups; three additional groups have conflicting labels. We isolated conflicts, six UAV images without corresponding positive mask pixels and their copies, and close background views at the split boundary. Table I gives the resulting fixed split.

Original video/session identifiers are unavailable. This is an inferred background-view holdout, with nearby facade views held out, zero cross-split exact-pixel duplicates, and no independent test split. Image lists, class mapping, grouping policy, and prepared-manifest digest are frozen. Conversion preserves pixels and native boxes; training applies aspect-ratio-preserving resizing and padding.

TABLE I
Fixed SIRST-UAVB Background-View Holdout

Quantity	Training	Validation
Images	1,438	505
Native UAV boxes	1,421	491
Bird boxes retained for analysis	925	227

III. Models and Experimental Settings

B0 retains prediction strides 8, 16, and 32 and the end-to-end output branch; its constructed training model has 2,504,190 parameters. Initialization transfers same-name, same-shape tensors from official YOLO26n layers 0–16; layers from 17 onward are random. B0 and P2 each copy 348 tensors. B1 copies 342, excluding its replaced shallow layer. This shared-prefix policy differs from full-model pretrained initialization.

All runs use one RTX 2080 Ti, Python 3.10.8, PyTorch 2.1.2+cu118, Ultralytics 8.4.184, image size 640, batch 4, seed 42, and four workers. B0's automatic optimizer resolved to AdamW; B1/P2 explicitly match learning rate 0.002, momentum 0.9, weight decay 0.0005, lrf 0.01, nbs 64, and warmup bias learning rate 0.0. The 300-epoch learning-rate and augmentation plan is retained, including mosaic, scaling, translation, and flipping.

B0 completed 92 full epochs and 317/360 batches of epoch 93 before its time callback stopped training. B1 and P2 completed exactly 92 epochs. B0/B1 selected epochs 81/68 by validation AP50:95 under the original AMP protocol. P2 keeps AMP optimization but changes validation and best selection to FP32 after the recovery checks below. No local training was performed.

IV. Results and Numerical Diagnosis

A. Historical B0 and B1 Results

Table II retains the original final-checkpoint revalidation at console precision. Precision and recall use the validator's operating point; AP50:95 averages IoU thresholds 0.50–0.95. These final values differ slightly from the selected in-training rows: B0 AP50/AP50:95 were 56.176%/23.061%, and B1 were 35.578%/14.111%.

NaN validation losses occur in B0 epochs 6, 7, 12, 31, 32, 36, 37 and B1 epochs 3, 6–12, 14, 16, 17. Training losses,

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TABLE II
HISTORICAL VALIDATION: 505 IMAGES, 491 UAV BOXES (%)

Model state	Precision	Recall	AP50	AP50:95
B0 partial epoch 93	72.4	48.3	52.6	20.6
B0 best (epoch 81)	66.5	49.2	56.2	23.2
B1 PConv best (epoch 68)	41.6	37.8	35.8	14.1

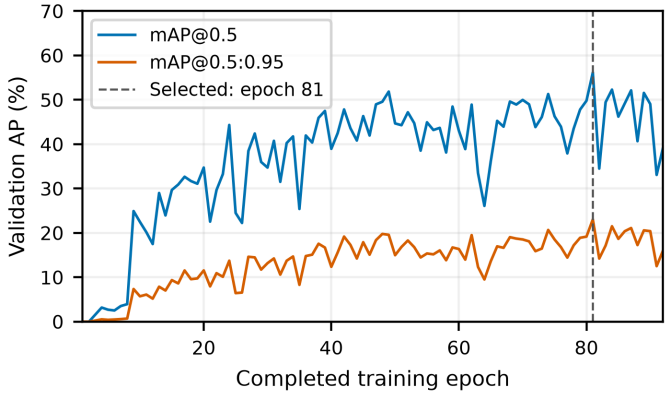


Fig. 1. B0 in-training AP through complete epoch 92; the dashed line marks epoch 81. Partial-epoch and post-training revalidation results are separate observations in Table II.

detection metrics, and saved floating-point weights remain finite. B1 training returned normally, but the strict all-history check rejected it and stopped the original queue. No rows were removed, replaced, or imputed.

B. Early EMA Replay and FP32 Recovery

The retained B0/B1 periodic checkpoints store zero-based epochs 30/10, corresponding to completed epochs 31/11, with EMA updates 834/384. Their EMA states are finite and B1 retains PConv. Each early EMA and final stripped best was evaluated in both precision modes on the same 505 images, batch 8, 64 batches, and fixed order, without gradients. Parameters and BN buffers were unchanged.

B0/B1 early AMP replay had 3/24 nonfinite batches; full FP32 replay had none. Both final bests were finite in both modes. The first captured top-level anomaly was `model.22.C3k2`. Recomputing FP32 loss from the same raw predictions of all 27 anomalous AMP batches remained nonfinite; full FP32 forward and loss were finite. This localizes an AMP forward-related path, not the internal failing operator or every historical NaN epoch. Both architectures exhibit it, so it cannot be attributed solely to PConv.

A project-scoped adapter gives the official training validator a read-only trainer view with AMP disabled, while preserving the actual trainer's AMP flag. FP32 preprocessing, forward, and loss passed on all four checkpoints and all 505 images with unchanged parameters/BN. Shared dependencies and historical CSVs were not modified.

C. P2 Recovery Experiment

P2 then completed 92 full epochs with all CSV values finite and 505-image FP32 validation each epoch. Final best revalidation after standard optimizer stripping used the same FP32 training-validator path and left model state unchanged. Box/class/L1 losses were 2.13623/1.87533/0.00130. Table III compares that outcome with independent FP32 revalidation of the existing B0/B1 bests.

TABLE III
FP32 TRAINING-VALIDATOR REVALIDATION (%)

Final best	Precision	Recall	AP50	AP50:95
B0 diagnostic control	67.918	49.287	57.339	23.087
B1 diagnostic control	40.057	38.086	35.523	14.026
P2 recovery	57.223	46.589	47.405	20.652

P2 does not exceed B0 here. P2 selects best using FP32 validation, whereas B0/B1 retain historical AMP-selected weights. Their new FP32 diagnostic metrics do not replace Table II; backend, batch, and preprocessing paths also differ from the historical standalone evaluation. Thus neither precision-only effects nor architecture-only gains can be inferred. Single-seed training and the inferred-view split further limit generalization.

D. Artifacts and Timing

Best/last files are each 5,388,158 bytes (B0), 5,394,626 (B1), and 5,622,388 (P2); final finite-state checks and remote/local SHA256 verification passed. P2's 29-file archive was verified before acknowledgement and platform shutdown. Raw histories and weights are preserved.

Historical B0 GPU validation reported 0.1/3.1/0.1 ms preprocessing/inference/postprocessing per image at batch 8. These offline framework timings do not measure decoding, camera input, display, or edge transfer. H10 conversion, Raspberry Pi execution, and real infrared-video accuracy and end-to-end latency remain unmeasured for these trained models.

V. CONCLUSION

Neither shallow PConv nor P2 improved observed AP over B0 in this single-seed study. Early-EMA replay reproduced AMP forward-related nonfinite values; FP32 validation enabled a separate P2 run with finite losses throughout 92 epochs. Historical NaNs remain recorded, and precision-dependent checkpoint selection limits direct comparison. Device conversion, video testing, and error analysis remain future work.

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